11.1 Influences on frequency, extent and severity of cyclones

Read the following information, answer the questions and complete the writing activity.

How does climate change increase the frequency and severity of tropical cyclones?

Climate change is a human-induced factor that is influencing the behaviour of tropical cyclones around the world. Human activities such as burning fossil fuels, deforestation and industrial processes release large amounts of greenhouse gases into the atmosphere. These gases trap heat, causing global temperatures to rise and leading to long-term changes in the climate system.

One of the most important ways climate change affects tropical cyclones is through warmer sea surface temperatures. Tropical cyclones form over warm ocean water, usually above 26.5°C. As oceans warm due to climate change, larger areas of the ocean meet this temperature threshold for longer periods of the year. This can increase the likelihood of cyclone formation and extend the length of the cyclone season in some regions, increasing the frequency of cyclones that form.

Climate change also increases the severity of tropical cyclones. Warmer oceans provide more energy to developing cyclones, allowing them to intensify more rapidly and reach higher wind speeds. Many recent cyclones, such as Cyclone Debbie in Australia and Cyclone Haiyan in the Philippines, intensified quickly before making landfall, causing widespread damage to infrastructure and communities.

In addition, warmer air temperatures caused by climate change allow the atmosphere to hold more moisture. This leads to heavier rainfall during cyclones, increasing the risk of river flooding, flash flooding and landslides. During Cyclone Debbie, extremely high rainfall totals caused major flooding across Queensland and northern New South Wales, damaging homes, roads and agricultural land long after the cyclone itself had weakened.

Rising sea levels are another important factor linked to climate change. As sea levels rise, storm surges become more destructive, as higher baseline sea levels allow seawater to travel further inland. This increases coastal flooding, erosion and saltwater contamination of freshwater supplies. Even cyclones of similar strength can cause greater damage today than in the past because sea levels are higher.

While climate change does not directly cause tropical cyclones, it intensifies the conditions that allow them to become more frequent and severe. As a result, communities are facing greater risks, higher recovery costs and more long-term impacts from atmospheric hazards.

Questions

1. Explain how warmer sea surface temperatures affect the frequency of tropical cyclones.

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2. Describe two ways climate change increases the severity of tropical cyclones.

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3. Using Cyclone Debbie as an example, explain how climate change influenced the impacts experienced in Australia.

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Writing activity

Respond to the following question.

Explain how ONE human-induced factor can increase the frequency and severity of atmospheric hazards.

It is encouraged to write two paragraphs, one on frequency and one on severity. A recommended writing structure is PEEIL: Point, Explain, Example, Impact and Link. The sentence starters and scaffold attached can be used as a guide.

Scaffold

PEEIL	Sentence starters	Planning notes
Point	Climate change affects how often atmospheric hazards occur by…	
Explain	Climate change causes global temperatures to rise, which leads to… As ocean temperatures increase, conditions become more suitable for…	
Example	For example, warmer sea surface temperatures above 26.5°C now occur… This means tropical cyclones can form…	
Impact	As a result, atmospheric hazards such as tropical cyclones occur… This increases the likelihood that communities will experience…	
Link	Therefore, climate change increases the **frequency** of atmospheric hazards.	

PEEIL	Sentence starters	Planning notes
Point	The severity of tropical cyclones is increased by climate change because…	
Explain	Warmer oceans provide more energy for cyclones, causing… In addition, warmer air can hold more moisture, which leads to…	
Example	For example, Cyclone Debbie produced extremely heavy rainfall, which… Rising sea levels also contributed to…	
Impact	This results in greater damage to homes, infrastructure and the environment. Communities experience more severe flooding, stronger winds and…	
Link	Overall, climate change makes atmospheric hazards more damaging.	